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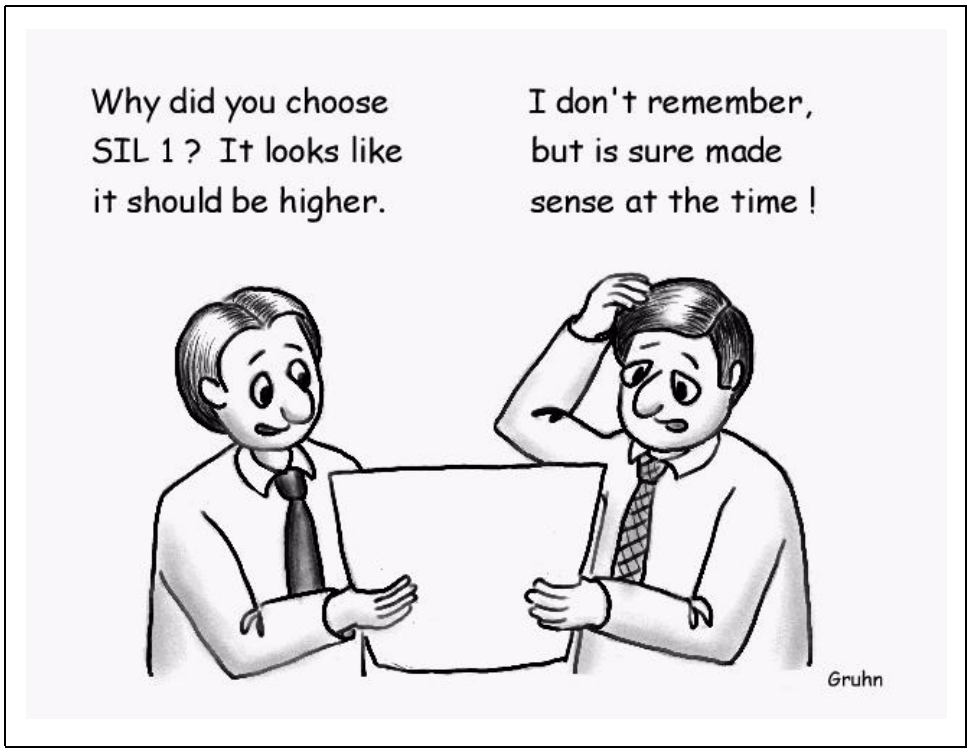
DETERMINING THE SAFETY INTEGRITY LEVEL (SIL)

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"The man who insists upon seeing with perfect clearness before deciding, never decides."

— H. F. Amiel



6.1 Introduction

Today's safety system standards are performance based, not prescriptive. They do not mandate a technology, level of redundancy, test interval, or system logic. Essentially they state "the greater the level of risk, the better the safety systems needed to control it." There are a variety of methods of evaluating risk. There are also a variety of methods of equating risk to the performance required from a safety system. One term used to describe safety system performance is safety integrity level (SIL).

Many industries have the need to evaluate and rank risk. Management decisions may then be made regarding various design options. For example, how remote, if at all, should a nuclear facility be to a large population zone? What level of redundancy is appropriate for a military aircraft weapons control system? What strength should jet engine turbine blades be for protection against flying birds? How long should a warranty period be based on known failure rate data? Ideally, decisions such as these would be made based upon mathematical analysis. Realistically, quantification of *all* factors is extremely difficult, if not impossible, and subjective judgment and experience may still be considered.

Military organizations were some of the first groups to face such problems. For example, when someone has to press the button that might start or stop World War III, one would like to think that the probability of the electronic circuitry working properly would be rated as something other than "high." The U.S. military developed a standard for categorizing risk: MIL-STD 882 "Standard Practice for System Safety" which has been adapted by other organizations and industries in a variety of formats.

Different groups and countries have come up with a variety of methods of equating risk to safety system performance. Some are qualitative, some more quantitative.

It's important to clarify what SIL is and what it is not. SIL is a measure of safety system performance. A system consists of a sensor, logic box, and final element(s). Therefore it is incorrect to refer to the SIL of an individual component of a system (e.g., a logic solver in isolation). A chain is only as strong as its weakest link. A logic solver could be rated for use in SIL 3, but if connected with non-redundant field devices with infrequent testing, the overall system may only meet SIL 1. There has been a push in industry for suppliers to use terms such as "SIL claim limit" and "SIL capability" in order to distinguish between component and system performance. It would be more appropriate to refer to the Probability of Failure on Demand (PFD) of an individual component. Unfortunately, PFD numbers are usually very small and referred to using scientific notation, which is difficult for some to relate to. Also, SIL is not a measure of process risk. It would be incorrect to say, "We've got a SIL 3 process."

6.2 Who's Responsible?

Determining SILs is mentioned in the various safety system standards. Many therefore assume the task falls upon the responsibility of the instrument or control system engineer. This is *not* the case. Evaluating the process risk and determining the appropriate safety integrity level is a responsibility of a multi-disciplined *team*, not any one individual. A control system engineer may (and should) be involved, but the review process will also require other specialists, such as those typically involved in a HAZard and OPerability study (HAZOP). Some organizations prefer to determine the SIL during the HAZOP. Others believe it's not necessary (or even desirable) to involve the entire HAZOP team and that the SIL determination should be performed as a separate study after the HAZOP. As a minimum, representatives from the following departments should participate in a SIL determination: process, mechanical design, safety, operations, and control systems.

6.3 Which Technique?

When it comes to hazard and risk analysis and determining safety integrity levels, there are no answers that could be categorized as either right or wrong. There are many ways of evaluating process risk, none more correct than another. Various industry documents describe multiple qualitative and quantitative methods for evaluating risk and determining SIS performance. [Ref. 2, 3, and 4] The methods were developed at different times by different countries. All methods are valid. Unfortunately, they may all also result in different answers.

Qualitative techniques are easier and faster to apply, but experience has shown they often provide conservative answers (i.e., high SIL requirements). This results in over-designing the safety functions and potentially unnecessary expenditures.

More quantitative techniques require greater effort in their development and usage, but experience has shown them to often provide lower SIL requirements. The difference in cost between a single SIL 2 versus a SIL 1 function can be tens of thousands of dollars. The difference between SIL 2 and 3 is even greater. Therefore, there may be significant cost savings associated with utilizing the more detailed quantitative techniques.

In general, if the use of qualitative techniques indicates no greater than SIL 1 requirements, continue their use. However, if the qualitative techniques indicate a significant percentage of SIL 2 or SIL 3 requirements, seriously consider the use of more quantitative techniques such as Layer Of Protection Analysis (LOPA). The overall cost savings that will result, especially over multiple projects, will easily justify their development and usage.

6.4 Common Issues

No matter what risk analysis method is chosen, several factors are common to all. For example, all methods involve evaluating the two components of risk—probability and severity—usually by categorizing each into different levels.

There are different hazardous events associated with each process unit and each event has its associated risk. Take for example a vessel where one is measuring pressure, temperature, level, and flow. The pressure measurement is probably meant to detect and prevent an over-pressure condition and explosion. This event would have a corresponding level of risk (i.e., probability and severity). Low flow might only result in a pump burning up. This would have a completely different probability and sever-

ity rating and, therefore, a different SIL requirement. High temperature might result in an off-spec product. This would also have a completely different probability and severity rating, and a different SIL requirement. What this means is one should *not* try and determine the SIL for an *entire* process unit or piece of equipment, but rather determine the SIL for *each safety function*.

6.5 Evaluating Risk

Risks are everywhere. Working in a chemical plant obviously involves risk, but then so does taking a bath. One involves more risk than the other, but there is a measure of risk in everything we do.

While zero injuries may be a goal for many organizations, it is important to realize that there is no such thing as zero risk. One is at risk merely sitting at home watching television (e.g., heart attack, flood, tornado, earthquake, etc.). So how safe should a process plant be? Should the risk of working at a chemical plant be equal to that of staying at home, or driving one's car, or flying in an airplane, or skydiving? All things being equal, the safer a plant is, the more expensive it will be. There has to be an economic consideration at some point. If the safety goal precludes the process from even being started, something obviously needs to be changed.

6.5.1 Hazard

The American Institute of Chemical Engineers (AIChE) definition of a hazard is "an inherent physical or chemical characteristic that has the potential for causing harm to people, property, or the environment. It is the combination of a hazardous material, an operating environment, and certain unplanned events that could result in an accident."

Hazards are always present. For example, gasoline is a combustible liquid. As long as there is no ignition source, gasoline may be considered relatively benign. Our goal is to minimize or eliminate hazardous events or accidents. We therefore don't store gasoline near ignition sources.

6.5.2 Risk

Risk is usually defined as the combination of the severity and probability of a hazardous event. In other words, how often can it happen, and how bad is it when it does. Risk can be evaluated qualitatively or quantitatively.

While many of the safety system standards would appear to focus on personnel risk, people are not the only thing at risk in process plants. For example, unmanned offshore platforms have a considerable amount of risk, even though there may be no personnel on the platform most of the time. Factors at risk are:

- **Personnel**. Both plant personnel and the nearby public.
- **Lost production downtime**. The loss associated with not being able to sell a product. This can often be the largest economic factor.
- **Capital equipment**. The costs associated with replacing damaged equipment.
- **Environment**. Polluting food supplies used by multiple countries or requiring the evacuation of an area is obviously something to be avoided.
- **Litigation costs**. Legal costs associated with injuries, deaths, and environmental losses can be significant.
- **Company image**. Bad press can put a company out of business.

Some of these items (e.g., lost production, capital equipment) may be quantified, some others (e.g., company image) may be more difficult to quantify.

Safety system performance levels can be assigned based on any of the risks listed above. For example, there are organizations that have safety integrity level (SIL) based on personnel risk, commercial integrity level (CIL) based on economic issues, and environmental integrity level (EIL) based on environmental factors.

6.5.3 Fatality Rates

The most obvious and serious risk to personnel is a fatality. How safe should personnel in or nearby a process plant be? Should working in a process plant be as safe as driving a car? In the U.S. there are approximately 45,000 fatalities every year due to automobile accidents. Would having a similar risk in the process industry mean the industry could kill 45,000 people every year? Obviously not! For one, not as many people work in process plants as drive automobiles. The issue is not total number of fatalities, but fatality rates.

There are two common methods of expressing fatality rates. One is the fatal accident rate, or FAR. It is the number of deaths per million person

hours of exposure. Another is the probability per unit of time. There are a number of sources that list fatality rates for various activities, from different industries, means of transportation, recreational activities, including voluntary as well as involuntary risks. Table 6-1 lists fatality rates for various activities in the U.K. [Ref. 1] Note that probabilities for two different activities may be the same, yet the FAR may be different, due to the difference in exposure hours. Also note that the numbers come from different sources and all of the underlying information may not be available, so there is not always the means of directly relating probabilities to rates, hence some of the fields in Table 6-1 are blank.

Table 6-1: Fatal Accident Rates (in the United Kingdom)

Activity	Probability (per year)	FAR
Travel		
Air	2×10^{-6}	
Train	2×10^{-6}	3 - 5
Bus	2×10^{-4}	4
Car	2×10^{-4}	50-60
Motorcycle	2×10^{-2}	500 - 1,000
Occupation		
Chemical Industry	5×10^{-5}	4
Manufacturing		8
Shipping	9×10^{-4}	8
Coal Mining	2×10^{-4}	10
Agriculture		10
Boxing		20,000
Voluntary		
The Pill	2×10^{-5}	
Rock Climbing	1.4×10^{-4}	4,000
Smoking	5×10^{-3}	
Involuntary		
Meteorite	6×10^{-11}	
Falling Aircraft	2×10^{-8}	
Firearms	2×10^{-6}	
Cancer	1×10^{-6}	
Fire	2.5×10^{-5}	
Falls	2×10^{-5}	
Staying at Home	1×10^{-4}	1 – 4

6.5.4 Risks Inherent in Modern Society

Just how safe is "safe"? Should working in a plant have the same level of risk as skydiving (which kills about 40 people per year in the U.S.)? Should working in a plant be as safe as driving your car? Or should it be as safe as flying in a plane, which is safer than driving a car by two orders of magnitude?

While the term FAR may be simple to understand and may represent a useful yardstick, many companies, especially in the U.S., are unwilling to put such targets in writing. Imagine walking into company XYZ's plush world headquarters office and on the wall in the reception area is a sign that reads, "We at XYZ consider it tolerable to kill 4 people per 100 million man hours." The lawyers would have a field day! However, as we shall see, some organizations *have* established such quantified risk targets.

People's perception of risk varies depending on their understanding or familiarity with the risk. For example, most people are familiar with driving. The perceived level of risk is relatively low to most people, even though approximately 45,000 people die every year in the U.S. alone due to traffic accidents. If a new chemical facility is being proposed nearby a residential area, the level of understanding of the residents regarding the chemical process will probably be low, and their discomfort, or perceived level of risk will no doubt be high, even if the process may have a very good historical safety record.

Perception of risk will also vary in proportion to the number of possible deaths associated with a particular event. For example, of the 45,000 traffic fatalities every year, the deaths usually occur one, or a few, at a time. Even with this surprisingly high number of deaths, there's little (if any) public outcry that something be done to lower this figure. Yet when there's an accident with a school bus involving injuries to many children there typically is an outcry. The same could be said about relatively high-risk sports such as skydiving, hang gliding, and ultralight aircraft. Although these sports involve relatively high risk, it's rare that one hears of multiple fatalities. The people involved also made their own conscious choice to partake in the activity and outsiders are generally not exposed to the risks. Accidents involving the chemical industry, however, frequently do involve multiple fatalities. Bhopal was the worst to date with over 3,000 deaths and 200,000 injuries. The overall risk associated with working in a chemical plant can be shown to be less than the risk of driving (at least in the U.S.), yet the public's *perception* of the risks of the two activities is typically reversed.

6.5.5 Voluntary vs. Involuntary Risk

There's a difference between voluntary and involuntary risks. Examples of voluntary risks would be driving a car, smoking cigarettes, and so on. Examples of involuntary risks would be having a chemical plant built near your home after you've lived there a number of years or secondary smoke from other peoples' cigarettes.

People can perceive similar risks differently. For example, the Jones own a house in the country. One day, company XYZ builds a toxic chemical plant nearby. After the plant is built, the Smiths buy a house next to the Jones. Both households face the same risk, but they'll probably each have a different perception of it. To the first couple (the Jones) who lived there before the plant was built, the risk is involuntary (although they obviously could move). To the second couple (the Smiths), who bought their home after the plant was built, it's voluntary.

People are usually willing to face higher voluntary risks than involuntary ones. For example, when one of the authors was younger he was willing to accept both the risks of riding a motorcycle in a large city and of skydiving. The risks were voluntary and he considered himself to be the only one at risk at the time. (One could argue the finer points of that.) As a married father, he no longer wishes to accept those risks (never mind the fact that he can no longer afford them).

Another factor involved in the perception of risk is control. For example, the wife of one of the authors does not like flying. (In fact, flying is the number two fear among Americans. Public speaking is number one!) Her stated reason for discomfort is that she doesn't feel "in control." When you're sitting behind the wheel in your car at a stop sign and a drunk driver plows into your car, you weren't in control then either. After all, no one goes out planning to have an accident.

6.5.6 Tolerable Levels of Risk

The concept of acceptable or tolerable levels of risk is not solely a technical issue; it involves philosophical, moral and legal matters. Deciding how safe is safe enough can't be answered by algebraic equations or probabilistic evaluations. Alvin Weinberg has termed these "transscientific questions," for they transcend science. [Ref. 7]

One issue that presents difficulties is trying to statistically estimate extremely unlikely events. Estimating very rare events, such as a severe chemical accident, cannot have the same validity as estimates for which

abundant statistics are available. Because the desired probabilities can be so small (e.g., 10-6 per plant per year), there is no practical means of determining the rate directly. One can't build ten thousand plants and operate them for one hundred years in order to tabulate their operating histories. Putting it a simpler way, measuring something that doesn't happen very often is difficult.

6.5.7 Tolerable Risk in the Process Industries

It's common to view personal risk in a subjective, intuitive manner. Many people won't consider driving a motorcycle, no matter how wonderful their biker friends say it may be. The author's wife who doesn't like flying believes it's all right for her family to drive to the airport in the same car but not to fly in the same airplane (even though she understands that flying is two orders of magnitude safer). Logic doesn't always apply when evaluating relative risk.

We should not, however, have the same subjective attitude about risks in the process industry. Usually, the people making the risk decisions (e.g., engineers) are not the ones who will be facing the eventual risk (e.g., workers or nearby residents). Although none of the more famous accidents in the process industry would ever be considered "acceptable," the companies involved did not go out of business the next day. Therefore, the losses must have been considered "tolerable." How many accidents might the industry be willing to consider tolerable? How many accidents must take place before there's public and political outcry?

There are approximately 2,300 petrochemical plants in the United States alone. If an average of one were to have a catastrophic event involving over 20 fatalities every year (which represents an individual plant risk of 1/2,300 per year), how long would it be before there was a public outcry and the government stepped in? What if such an accident only happened once every ten years (1/23,000 per year)? There is no such thing as zero risk, but it's very difficult to decide what particular level should be considered "tolerable."

The relevant statistics can be rather confusing. An individual risk of 1/2,300 per year means that out of 2,300 plants, on average, one might go "boom" every year. It's important to realize, however, that you can't predict *which* plant and you can't predict *when* one will go "boom." But since people don't build 2,300 plants all at once, or live next to 2,300 plants, they want to know the risk of the one plant they're associated with. The risk for an individual plant remains the same, 1/2,300 per year. However, some are just not comfortable with such a number. Some twist things around a

bit and say the risk of an accident is "once every 2,300 years." This causes even more confusion. Some then assume it will be 2,300 years before there's an accident and, therefore, they have nothing to worry about. Nothing could be further from the truth. For example, approximately 1 in 4,000 people in the U.S. die in a car crash every year. If you go to a sports event with 4,000 people present, one can make the assumption that someone will die in an automobile accident within the next 365 days. However, you can't predict *which* person and you can't predict which *day*. The error associated with inverting the number and stating that you'll live 4,000 years before you die in a car crash should now be obvious.

Deciding what level of risk is tolerable could be compared to choosing your weapon in Russian roulette—how many barrels do you want in your gun? Would you choose an automatic pistol that always had a round in the chamber? (I hope not, although there have been such Darwin award winners!) Or would you choose a revolver with six chambers? What if you could choose a gun with fifty barrels or one with five thousand barrels? Obviously, the more barrels there are, the lower the risk of hitting the bullet when you pull the trigger. Either way, you have to play the game. You do, however, at least get to choose your weapon. There is no such thing as zero risk.

One in 2,300 per year, or the incorrect reciprocal of 2,300 years between accidents, may initially sound so remote as to be of no concern to most people. A more intuitive answer might initially be fifty years. (The reasoning being that someone's working life is fifty years, and they don't want anything to happen during *their* life.) But how long will the plant actually be around? Let's say 25 years. So take a gun with fifty barrels and pull the trigger once a year. What's the likelihood of hitting the bullet? 50%. (Although based on another set of assumptions and simplifications, the answer is 40%. Even the statistics involved are not inherently obvious!)

Would you want to have your name on the drawings of a plant if you knew there was a 50% chance of a catastrophic accident happening during the life of the plant? Probably not.

What if instead of 50 years, you choose 500 years? Then the risk would now be 25/500, or a 5% chance. Should that be tolerable? What about 5,000 years? Now it becomes 0.5%. There is no such thing as zero risk, but just how low must one go for the risk to be considered tolerable? That's the proverbial $64,000 question to which there is no scientific answer.

U.S. industry cannot afford to blow up an entire plant, involving dozen of fatalities, once every year (1/2,300). The negative press and public outcry would be ruinous. After all, OSHA stepped in and produced 29 CFR

1910.119 after several major Gulf Coast accidents happened over approximately a ten-year period. A serious accident once every ten years *might* be viewed as tolerable (1/23,000). Risk targets in the range of 1/10,000 have been documented. In fact, in The Netherlands, the government even publishes what it considers to be tolerable fatality rates. Switzerland and Singapore have done the same. Fatality rates are typically in the range of 1/1,000,000 per year. [Ref. 8]

6.6 Safety Integrity Levels

The greater the level of process risk, the better the safety system needed to control it. Safety integrity level is a measure of safety system performance. It is *not* a direct measure of process risk.

Standards and guidelines have categorized four overall safety integrity levels. Note that previous U.S. process industry documents only went up to SIL 3. The basis for the performance goals listed in Tables 6-2 and 6-3 have been a subject of controversy. In other words, how were these tables "calibrated"? Perhaps the numbers should be shifted up a row, or perhaps down a row. Why start at 90% and not 80%? Suffice it to say that the tables were accepted by different industry groups and appear in all the standards.

Table 6-2: Safety Integrity Levels and Required Safety System Performance for Low Demand Mode Systems

Safety Integrity Level (SIL)	Probability of Failure on Demand (PFD)	Safety Availability (1-PFD)	Risk Reduction Factor (1/PFD)
4	.0001 - .00001	99.99 – 99.999%	10,000 – 100,000
3	.001 - .0001	99.9 - 99.99%	1,000 - 10,000
2	.01 - .001	99 - 99.9%	100 - 1,000
1	.1 - .01	90 - 99%	10 - 100

Table 6-3: Safety Integrity Levels and Required Safety System Performance for Continuous Mode Systems

Safety Integrity Level (SIL)	Dangerous Failure per Hour
4	$\geq 10^{-9} < 10^{-8}$
3	$\geq 10^{-8} < 10^{-7}$
2	$\geq 10^{-7} < 10^{-6}$
1	$\geq 10^{-6} < 10^{-5}$
Notes: 1. Field devices are *included* in the above performance requirements. 2. Refer to Section 3.3.2 of this text for a discussion of low vs. continuous mode systems	

Different people use different terms to describe system performance. A very common term, especially among control system vendors, is the generic term 'availability'. The ISA SP84 committee purposely used a different term since the safety standards are referring to a *completely* different type of performance. This subject is discussed in more detail in Chapter 8. Safety availability can still be a confusing term as most figures are so close to 100%. For example, do you consider there to be a significant difference between 99% and 99.99%? The difference could be viewed as being less than 1%.

The compliment of safety availability is referred to as the probability of failure on demand (PFD). Now the numbers are so small that one must use scientific notation. How many plant managers will understand when their control system engineer tells them that their safety instrumented function has a PFD of 4.6×10^{-3}?

The reciprocal of PFD is called the risk reduction factor (RRF). The benefit to this term is that the difference between numbers is easier to see. For example, the difference between a risk reduction factor of 100 and 10,000 is obviously two orders of magnitude. Actually, the difference between a safety availability of 99% and 99.99% is also two orders of magnitude, although most don't realize it initially.

The following sections describe the various methods of determining the required safety integrity level based on process risk issues. The methods are described in both the IEC 61508 and 61511 standards and other texts. Please keep in mind that no SIL determination methodology is better or more correct than another. They were simply developed by different groups around the world at different times. Similarly, a Peugeot is probably no better than a Ford. Both are simply cars developed by different countries. No doubt national pride will come into play when qualifying each, but both serve their purposes equally well.

6.7 SIL Determination Method #1: As Low As Reasonably Practical (ALARP)

The English developed the theory of ALARP which is considered a *legal* concept. While not a method for determining safety integrity levels directly, it is a concept that can still be applied in this area. The idea hinges on three overall levels of risk and the economics associated with lowering the risk. The three overall levels of risk are defined as "unacceptable", "tolerable", and "broadly acceptable" as shown in Figure 6-1.

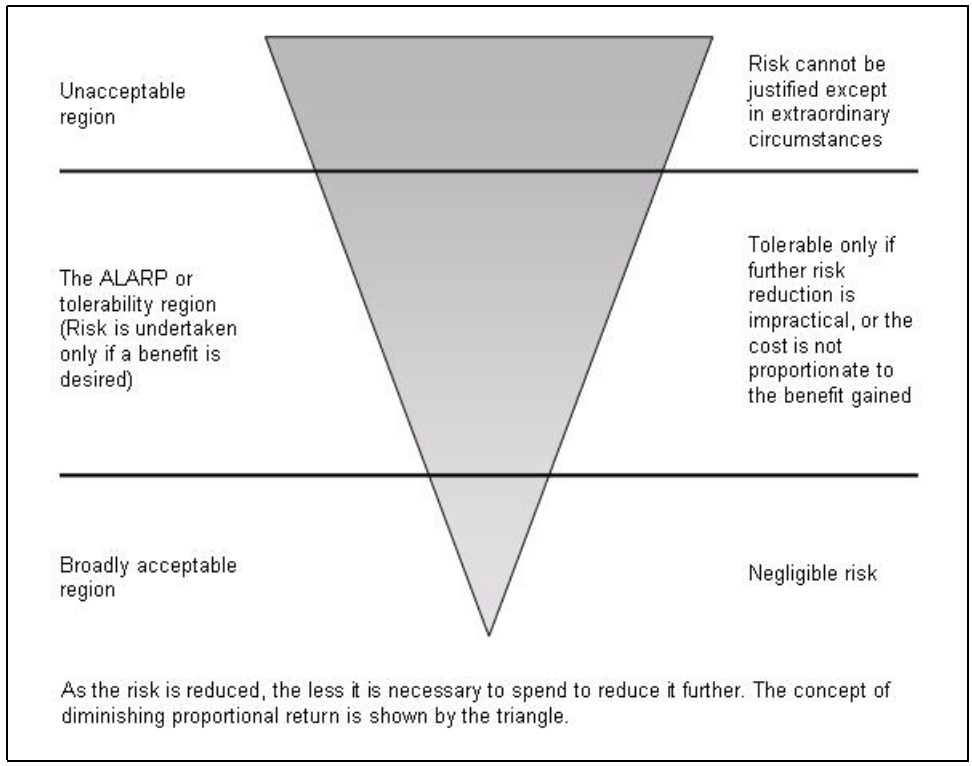


Figure 6-1: As Low As Reasonably Practical (ALARP) Principle

"Unacceptable" risks are viewed as so high as to be intolerable. For example, if a risk analysis showed the probability of destroying a large, manned, deepwater offshore platform at once per year, it's doubtful the design would proceed. Something must be done to lower the risk to a more tolerable level, regardless of cost.

"Broadly acceptable" risks are ones that are so low as to not be of major concern. For example, it is possible for a refinery to be hit by a meteor or terrorist airplane attack. However, building a missile defense system around every process plant would hardly be considered warranted or economically feasible.

In between these two extremes are risks that are considered "tolerable" if the positive benefit is seen to outweigh the potential negative impact. Driving a car is an excellent example. The fact that approximately 45,000 people in the U.S. die in auto accidents every year does little to deter the rest of us from driving.

While such decisions are primarily economic, other factors do influence company judgments. For example, no nuclear power plants have been built in the U.S. for over two decades. Similarly, few grassroots chemical

plants are being built in the U.S., no matter how safe they may be shown to be, or no matter how positive the economic factors for a community may be. The combined power of Not In My Back Yard (NIMBYs) has influenced many companies to build their facilities overseas where public, political, legal, and economic conditions are more favorable.

6.8 SIL Determination Method #2: Risk Matrix

The risk matrix found in many standards, recommended practices, and internal company procedures has many similarities to a U.S. military standard (MIL STD 882). Both categorize frequency and severity of a hazardous event using multiple qualitative levels. The risk matrix common in the process industry, however, often adds a third axis to account for additional safety layers.

6.8.1 Evaluating the Frequency

The frequency or probability of an event may be ranked from low to high, improbable to frequent, or whatever term may be appropriate. This may be for a single item, or for a group of items, a single process unit, or an entire plant. Levels may be ranked qualitatively or quantitatively as shown in Table 6-4. If quantitative values are chosen, it's suggested that they differ by at least an order of magnitude, since the safety integrity levels differ by single orders of magnitude.

Table 6-4: Risk Frequency (Example Only)

Level	Descriptive Word	Qualitative Frequency	Quantitative Frequency
5	Frequent	A failure that can reasonably be expected to occur more than once within the expected lifetime of the plant	Freq > 1/10 per year
4	Probable	A failure that can reasonably be expected to occur within the expected lifetime of the plant	1/100 < Freq < 1/10 per year
3	Occasional	A failure with a low probability of occurring within the expected life of the plant.	1/1,000 < Freq < 1/100 per year
2	Remote	A series of failures, with a low probability of occurring within the expected life of the plant	1/10,000 < Freq < 1/1,000 per year
1	Improbable	A series of failures, with a very low probability of occurring within the expected life of the plant	Freq < 1/10,000 per year

6.8.2 Evaluating the Severity

Severity may also be categorized according to the different factors at risk; people, capital equipment, production, etc. The numbers shown in Tables 6-4 and 6-5 are *examples only* and are *not* intended to represent, or even imply, any kind of recommendation. The monetary values shown in Table 6-5 are also examples only and need to be "calibrated" for each organization. For example, what may be a catastrophic financial loss to one organization may be a relatively trivial loss to another.

Table 6-5: Risk Severity (Example Only)

Level	Descriptive Word	Potential Severity/Consequences		
		Personnel	Environment	Production/ Equipment
V	Catastrophic	Multiple fatalities	Detrimental offsite release	Loss > $1.5M
IV	Severe	Single fatality	Non-detrimental offsite release	Loss between $1.5M and $500K
III	Serious	Lost time accident	Release onsite - not immediately contained	Loss between $500K and $100K
II	Minor	Medical treatment	Release onsite - immediately contained	Loss between $100K and $2,500
I	Negligible	First aid treatment	No release	Loss < $2,500

6.8.3 Evaluating the Overall Risk

The two sets of numbers may then be combined into an X-Y plot as shown in Table 6-6. The lower left corner represents low risk (i.e., low probability and low frequency), the upper right corner represents high risk (i.e., high probability and high frequency). Note that defining the borders between categories is rather subjective. One may also use more than three overall categories.

These risk levels are similar to the ALARP concept. If high risk is identified, process changes may be called for, whether the facility is new or existing. If low risk is identified, no changes may be warranted. If moderate risk is identified, additional safety layers or procedures may be necessary. Note that these are merely examples.

Table 6-6: Overall Risk (Example Only)

	Frequency				
Severity	**1**	**2**	**3**	**4**	**5**
V	1-V	2-V	3-V	4-V	5-V
IV	1-IV	2-IV	3-IV	4-IV	5-IV
III	1-III	2-III	3-III	4-III	5-III
II	1-II	2-II	3-II	4-II	5-II
I	1-I	2-I	3-I	4-I	5-I

High Risk

Low Risk

Medium Risk

The two dimensional matrix can be used to evaluate overall risk. It should *not*, however, be used to determine safety integrity levels as there are additional process design issues that should be accounted for, such as additional safety layers, as covered in the next section.

6.8.4 Evaluating the Effectiveness of Additional Layers

A third axis appears in a variety of standards and guidelines which does not appear in the military standard (see Figure 6-2). This third axis is designed to take into account the additional safety layers commonly found in the process industries. Note that the Z axis is labeled "quantity and/or effectiveness of additional layers." This refers to layers *out*side of the safety instrumented system shown in the "onion diagram" (see Figure 4-2). In other words, if the safety system under consideration were to fail, might any other layer(s) be able to prevent or mitigate the hazardous event? Note that the basic process control system, alarms, and operators are already assumed to be in place and should *not* be accounted for in the Z axis. If, for example, a redundant process control system were being considered, it might be possible to consider a lower frequency/probability on the X axis. This would essentially have the same overall effect of lowering the required safety integrity level of the safety system.

The numbers shown in the squares in Figure 6-2 represent safety integrity levels and are merely examples. Which number is placed in which box is *not* consistent across industry and needs to be settled on within each organization. For example, previous U.S. standards did not recognize SIL 4. Just because SIL 4 is now defined in more recent standards does not necessarily mean that organizations need to modify previous graphs, especially if they felt that they did not even have any SIL 3 requirements.

One difficulty associated with Figure 6-2 (as well as Table 6-4) is that frequencies should be chosen assuming the safety layers are *not* in place. If the hazardous event under consideration were an overpressure of a vessel resulting in an explosion, the historical records of vessel explosions would already take into account safety layers such as safety systems, pressure relief valves, rupture discs, etc. Estimating what hazardous event frequencies *might* be with*out* these layers proves to be a difficult exercise.

As an example, suppose the hazardous event under consideration were an overpressure condition resulting in an explosion of a vessel. If the area were normally manned, there might be multiple fatalities. This might result in the highest severity rating. Historical records of vessel explosions might indicate a very low frequency, although the records already account for various safety layers. Again, this is merely a hypothetical example. If there were no additional safety layers beyond the safety instrumented system, the Z axis cannot be utilized. This would result in a SIL 3 requirement as shown in Figure 6-2. However, if there are additional layers, such as a relief valve, one could utilize the Z axis and lower the design requirements of the safety system under study from SIL 3 to SIL 2. This would be accounting for the relief valve as a single layer. Note, however, that the Z axis can also be used to account for the *effectiveness* of additional layers. A quantitative analysis of a relief valve might show it to have an effectiveness equal to two layers. (This concept will be covered in more detail in the layer of protection analysis [LOPA] section.) This means the required safety integrity level might be lowered from SIL 3 down to SIL 1.

One concern using this safety matrix method has to do with the qualitative judgments involved. Every company can take the liberty of defining the levels and rankings a bit differently. Occupational Safety and Health Administration (OSHA) representatives have admitted that they are not process experts. They are not about to tell a user that they should have selected SIL 3 instead of SIL 2 for a particular function. They just want to see the paperwork trail your company has left behind and the decision process of how you got there. The fact that different locations within the same company can define all of this differently does not really matter (at least to the standards bodies), even though it is naturally cause for concern. The industry simply has not reached, and may never reach, consensus on all of the subjective decisions and ranking involved.

There's a simple yet rather crass thought to keep in the back of your mind during this whole process, "How will we defend this decision in a court of law?" As scary as that thought may seem, it sums it up rather well. Think about it. If there were an accident and people were injured, the case would most likely go to court. Someone somewhere will need to be able to defend their design decisions. If asked why a certain decision was made

and the response is, "Well, we weren't sure what to do, but that's what our vendor recommended." the court's response may naturally be less than forgiving.

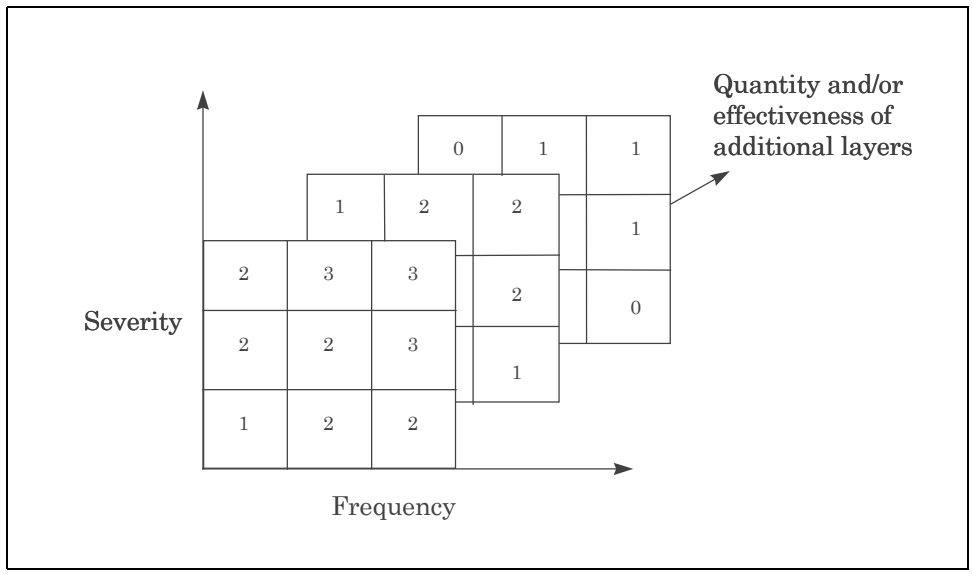


Figure 6-2: 3-Dimensional Risk Matrix for Selecting SIL

6.9 Method #2: Risk Graph

The Germans and Norwegians developed different, yet similar, techniques for determining safety integrity levels (see Figure 6-3). The wording used in the figure is purposely vague and open to interpretation. Each organization needs to define the boundaries/classifications more precisely. The standards are simply trying to show the methodology, not a working example that might be misused.

Figure 6-3 was intended for the risk to personnel. Similar graphs can be developed for risks associated with equipment, production losses, environmental impact, etc.

One starts at the left end of the chart. First, what are the consequences to the people involved for the hazardous event being considered? What if there are four fatalities? Would that be considered "several deaths" or "many deaths"? Again, each organization is expected to define the boundaries more precisely.

Next, what's the frequency and exposure of the personnel to the particular hazardous event? Keep in mind that even though a process might be con-

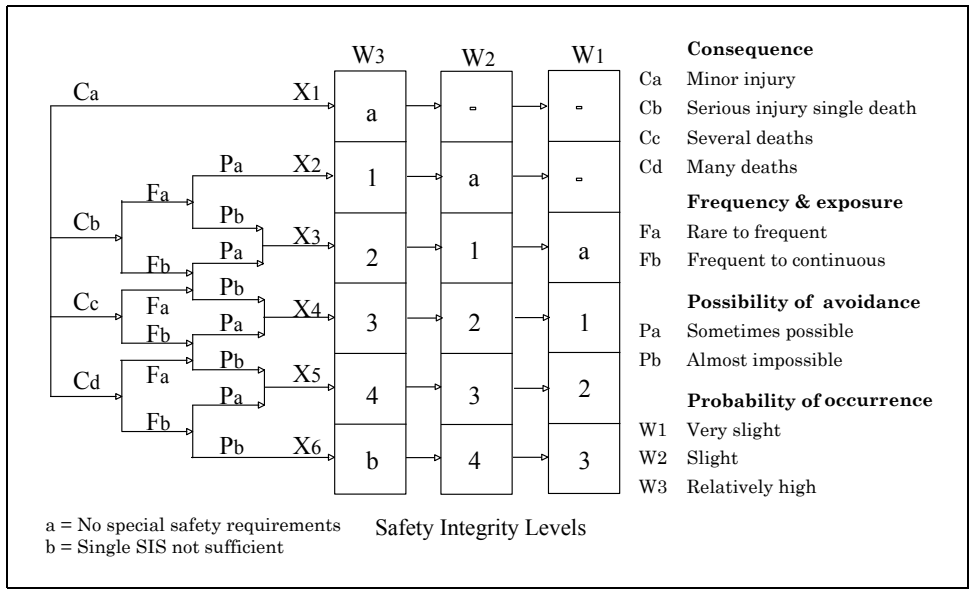


Figure 6-3: Risk Graph

tinuous, the personnel might only be in the particular area 10 minutes out of the day, so their *exposure* would *not* be continuous. One should also consider the increased manning that might occur during a hazardous event buildup. Frequent to continuous exposure may be rare. One obvious example might be living and working in a city with a nuclear power plant.

Next, one considers the possibility of avoiding the accident. There are multiple factors one may account for here. For example, what is the "process safety time," which is essentially defined as the time it takes for a hazardous event to occur if all control were lost? Is the process reaction time slow enough for the people to take action, or are the process dynamics so fast that operator intervention might not be possible? Do the operators have local indications to know what is going on in the process (so they can take corrective action)? Are personnel adequately trained to know what to do when an abnormal situation develops (so they may attempt to prevent it)? Are evacuation routes accessible in order to vacate the area in case the hazardous event were to occur? The differences between "sometimes possible" and "almost impossible" should be more clearly defined by each organization.

Finally, one considers the probability of the event actually occurring, again assuming the safety system were not in place. This primarily has to do with the additional safety layers and historical records of similar incidents. In other words, considering all of the other independent safety layers, what's the actual likelihood of this particular event happening?

The differences between "very slight," "slight," and "relatively high" should be more clearly defined by each organization.

One can see that all of this is rather subjective. One may also question the calibration of Figure 6-3. Who's to say this chart should also not be shifted up or down one row? No doubt the Europeans had some sort of rationale in its development.

6.10 Method #3: Layer of Protection Analysis (LOPA)

Some engineers are not comfortable with the purely qualitative methods outlined above for determining SIL. The repeatability of the methods has been questioned by some. For example, if you give the same problem to five different design groups and two groups come up with SIL 1, two groups with SIL 2, and one group with SIL 3, what might you conclude? Such differences would result in vastly different SIS designs. Are some of the groups "right" and others "wrong"? Actually, defining the boundaries and training the teams properly should make the overall determination process more consistent and repeatable.

Some are more comfortable with a more quantitative method. For example, if your company does not want the hazardous event to happen more often than once in a thousand years, and experience shows there to be a process demand on average once a year, then it's a relatively straightforward matter to determine the performance required of your safety layers.

Two AIChE CCPS textbooks describe LOPA. [Ref. 3 and 5] The onion diagram shown in Figure 4-2 illustrates how there are multiple layers of protection in any process plant. Each layer will have its own associated level of performance, or risk reduction. (Note that determining the actual performance of any one layer is the subject of Chapter 8.) If one knows a) the overall safety target (i.e., the tolerable risk level), b) what level of risk one is starting from (i.e., the initiating event frequency), c) how many safety layers are in place to prevent or mitigate the event, and d) the level of performance of each layer, one can calculate and determine if the overall target level of risk is being achieved. The concept is relatively easy to understand and is shown in Figure 4-3. However, coming up with numbers for all of the above is a bit of a challenge.

6.10.1 Tolerable Risk

The first step involves determining a tolerable level of risk. This topic was discussed earlier in Section 6.5. The legal system in the U.S. makes determining and documenting such numbers problematic; however, it has been

done. Also, certain governments around the world have documented such a number for industry to use. Table 6-7 (from Ref. 5) lists examples of yearly tolerable fatality probabilities. Company names have been removed from this table. Similar tables can be developed for tolerable probabilities for different size losses due to the impact of environmental, lost production downtime, capital equipment, and other events.

Table 6-7: Sample Tolerable Risk Probabilities

	Max tolerable risk for workforce (all scenarios)	Negligible risk for workforce (all scenarios)	Max tolerable risk for public (all scenarios)	Negligible risk for public (all scenarios)
Company A	10^{-3}	10^{-6}	-	-
Company B	10^{-3}	10^{-6}	-	-
Company C	3.3×10^{-5}	-	10^{-4}	-
Company D	2.5×10^{-5} (per employee)	-	10^{-5}	10^{-7}
Typical (one scenario)	10^{-4}	10^{-6}	10^{-4}	10^{-6}

6.10.2 Initiating Event Frequencies

The next step involves determining initiating event frequencies or probabilities. These can either be external events (e.g., a lightning strike) or failure of one of the layers (e.g., a control system valve fails open leading to a hazardous event development). Numbers need to be determined for any event under consideration. Table 6-8 (from Ref. 5) as an example. Numbers may be determined based on historical records or failure rate data (which is one form of historical record).

Table 6-8: Sample Initiating Event Frequencies

Initiating Event	Frequency (per year)
Gasket / packing blowout	1×10^{-2}
Lightning strike	1×10^{-3}
BPCS loop failure	1×10^{-1}
Safety valve opens spuriously	1×10^{-2}
Regulator failure	1×10^{-1}
Procedure failure (per opportunity)	1×10^{-3}
Operator failure (per opportunity)	1×10^{-2}

Note that while operating and maintenance procedures may be considered by some to be a safety layer, it is recognized that errors will be made and procedures will be violated. Human issues such as these may in *some* cases be considered initiating events, in others they may be prevention layers.

6.10.3 Performance of Each Safety Layer

The next step is determining the level of performance of each safety layer. There are rules on what may be considered an independent protection layer (IPL). For example:

1. **Specificity**: An IPL is designed solely to prevent or to mitigate the consequences of one potentially hazardous event. Multiple causes may lead to the same hazardous event. Therefore, multiple event scenarios may initiate action of one IPL.
2. **Independence**: An IPL is independent of the other protection layers associated with the identified danger. The failure of one layer will not prevent another layer from working.
3. **Dependability**: It can be counted on to do what it was designed to do. Both random and systematic failures modes are addressed in the design.
4. **Auditability**: It is designed to facilitate regular validation of the protective functions. Proof testing and/or maintenance is necessary.

Table 6-9 (from Ref. 5) shows as an example of performance that may be assigned to different levels. The probabilities are primarily based on failure rates and test intervals. (Chapter 8 shows examples of how such probabilities may be calculated.)

Table 6-9: Sample PFDs for Different Independent Protection Layers

Passive Independent Protection Layers	Probability of Failure on Demand
Dike	1×10^{-2}
Fireproofing	1×10^{-2}
Blast wall / bunker	1×10^{-3}
Flame / detonation arrestors	1×10^{-2}
Active Independent Protection Layers	
Relief valve	1×10^{-2}
Rupture disk	1×10^{-2}
Basic Process Control System	1×10^{-1}

Considering human action as a safety layer can be difficult and controversial since human errors are well noted. In order to account for human as a safety layer, all of the following conditions must be met:

- their intended procedures must be documented,
- personnel must be trained on the specific procedures,
- personnel must have sufficient time to act (e.g., 15 minutes),
- the use of procedures must be audited

Even with the above conditions satisfied, a probability limit (cap) on human error of 1×10^{-1} is a reasonable. In other words, claiming that a person will respond correctly in a high stress emergency situation more often than 9 out of 10 times (even with the above four cases satisfied) may be considered unrealistic. The use of specialized techniques for evaluating human reliability would be required in order to make such a claim, and such techniques are rarely used in the process industries.

Table 6-9 is obviously just a partial listing of potential passive and active safety layers. A fire and gas system may be one of the safety layers and could have its performance quantified (which is the topic of Chapter 8). In many cases the performance of F&G systems will be limited to a PFD of approximately 1×10^{-1}. If the F&G system is to be considered a safety system and have an integrity level assigned (according to the procedures outlined in this chapter), then its performance (PFD) requirements may have to be considerably greater.

6.10.4 Example Using LOPA

The following example is a slight modification of one covered in Reference 5. Figure 6-4 shows the sample process. A vessel is used to store hexane, a combustible material. The level in the vessel is controlled by a level controller which operates a valve. If the vessel is overfilled, hexane will be released through a liquid vent and be contained within a dike. A hazard analysis was performed and it was determined that the level controller might fail, liquid might be released outside of the dike, an ignition source might ignite the hexane, and there might be a possible fatality. The organization wanted to determine if the existing facility would meet their corporate risk criteria, or if any changes were required (such as adding a stand-alone safety system), how extensive the changes would need to be.

The organization established a yearly tolerable risk limit for a fire of 1E-4 and 1E-5 for a fatality. The initiating event for this scenario would be a failure of the control system, which was estimated at 1E-1. The only existing safety layer would be the dike, which had an estimated PFD of 1E-2.

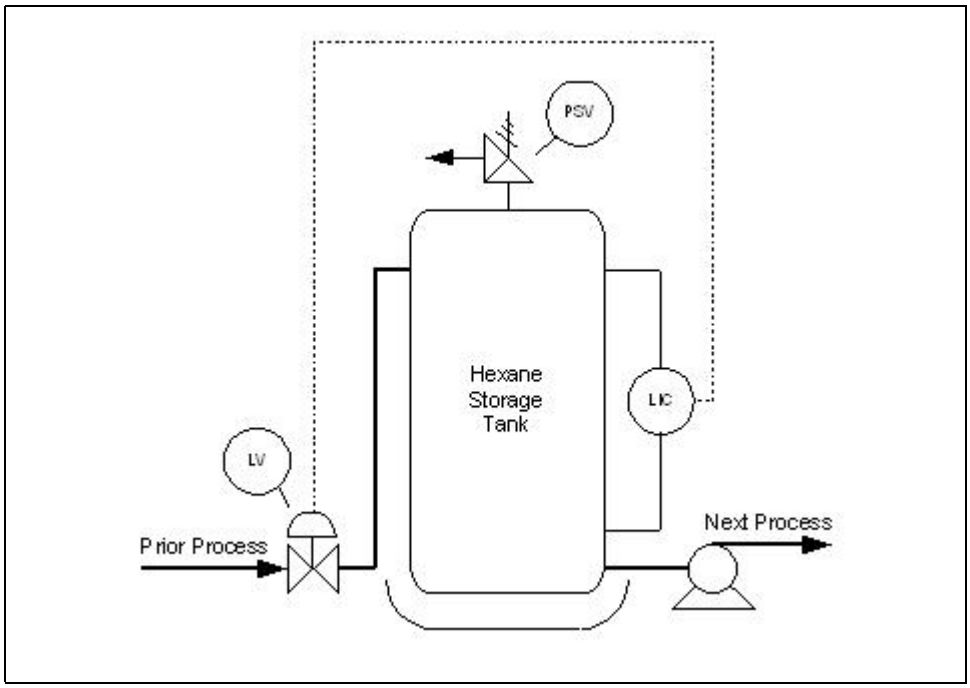


Figure 6-4: Sample Process for LOPA Example

Alarms and operator action were *not* accounted for because, in this instance, the control system was the initiating event, therefore no alarms would be generated. The organization took a conservative view that if material was released outside of the dike, the likelihood of it finding an ignition source would be 100%. However, the area was not always manned. The probability of someone being in the area was estimated at 50%. The probability of someone in the area actually being killed by a fire, as opposed to merely injured, was estimated at 50%.

Figure 6-5 shows an event tree for this scenario. The probability of a fire is represented by the combination of probabilities of the bottom three rows, which amounts to 1E-3 (0.1 x 0.01 x 1.0). The probability of a fatality is represented by the bottom row, which amounts to 2.5E-4 (0.1 x 0.01 x 1.0 x 0.5 x 0.5).

Knowing that the corporate risk target for a fire is 1E-4, we can see that the risk target is not being met by a factor of 10 (1E-3 / 1E-4). Knowing that the corporate risk target for a fatality is 1E-5, we can see that the risk target is not being met by a factor of 25 (2.5E-4 / 1E-5). Therefore, the existing design does *not* meet either corporate risk target and a change is warranted. One possible solution would be to install a separate high level shutdown function. Such a function would need to reduce the risk by at least a factor of 25 in order to meet the overall criteria. A risk reduction

factor (RRF) of 25 falls within the SIL 1 range (10-100) as shown in Table 6-2. However, obviously not *any* SIL 1 system will do. The new system must meet a Risk Reduction Factor of at least 25. Figure 6-6 shows a sample worksheet for documenting this scenario.

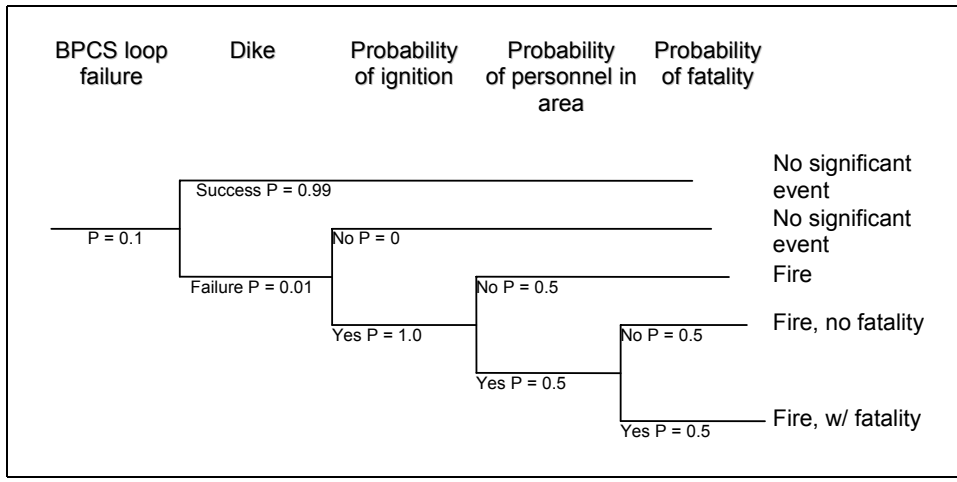


Figure 6-5: Event Tree for LOPA Example

Summary

Many industries have a need to evaluate and rank risk and match it to safety system performance. There are a variety of techniques that can be used ranging from qualitative to quantitative. All methods involve evaluating at least the two components of risk; probability and severity. No method is more accurate or better than another. All are valid techniques developed in different countries at different times.

The task of performing hazard and risk analyses and matching risk to safety system performance does not fall solely on the instrument or control system engineer. It's a task that must be carried out by a multi-disciplined team. Also, one determines the SIL (safety integrity level) for each safety instrumented function (SIF). It's incorrect to refer to the SIL of an entire process unit or piece of equipment.

LOPA Worksheet

Scenario Number:	Equipment Number:	Scenario Title: Hexane Surge Tank Overflow. Spill not contained by the dike.	
Date:	**Description**	**Probability**	**Frequency (per year)**
Consequence Description / Category	Release of hexane outside the dike due to tank overflow and failure of dike with potential for ignition and fatality		
Risk Tolerance Criteria (Category or Frequency)	Maximum tolerable risk of serious fire Maximum tolerable risk of fatal injury		$< 1 \times 10^{-4}$ $< 1 \times 10^{-5}$
Initiating Event (typically a frequency)	BPCS loop failure		1×10^{-1}
Enabling Event or Condition		N/A	
Conditional Modifiers (if applicable)			
	Probability of ignition	1	
	Probability of personnel in area	0.5	
	Probability of fatal injury	0.5	
	Other	N/A	
Frequency of Unmitigated Consequence			2.5×10^{-2}
Independent Protection Layers			
	Dike (existing)	1×10^{-2}	
Safeguards (non-IPLs)			
	Human action not an IPL as it depends upon BPCS generated alarm. (BPCS failure considered as initiating event.)		
Total PFD for all IPLs		1×10^{-2}	
Frequency of Mitigated Consequence			2.5×10^{-4}
Risk Tolerance Criteria Met? (Yes/No): No. SIF required			
Actions Required:	Add SIF with PFD of at least 4×10^{-2} (Risk Reduction Factor > 25) Responsible Group / Person: Engineering / J.Q. Public, by July 2005 Maintain dike as an IPL (inspection, maintenance, etc.)		
Notes:	Add action items to action tracking database		

Figure 6-6: Sample LOPA Example

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